Chapter 1

Introduction

1.1 Rationality Problems (TODO)

One of the oldest and most fruitful circle of questions in algebraic geometry

A fundamental idea in algebraic geometry is that it is very useful and rewarding to study algebraic varieties up to birational equivalence. Recall that two irreducible varieties X and Y are said to be birationally isomorphic if there are Zariski open subsets U X and V Y such that U = V . This is a natural concept to consider since many natural geometric constructions – for example, blowing up a subvariety – preserve the birational equivalence class of a variety. In the late 1800s and early 1900s, there were many significant advances in our understanding of birational transformations and various birational invariants of algebraic surfaces, which led to the Enriques-Kodaira classification of compact algebraic surfaces (see for instance [11]). In higher dimension, the past 50 years have witnessed vast progress in birational geometry, culminating in the near completion of the minimal model program ([12], [27]). From the birational viewpoint, the simplest varieties are those that are rational. By definition, an n-dimensional variety is rational if it is birationally equivalent to Pn. In dimension 2, the situation is fairly well-understood. For example, an algebraic surface is rational if and only if it is the blow-up of P2, P1 × P1, or a Hirzebruch surface. However, the situation in dimension 3 is already much more subtle (see [30], [3], and [18]), and has been the subject of current active research in the past 20 years.

1.2 Unirationality: examples, obstructions and open problems

However, when varieties are provably irrational, there remain interesting gradations in their complexity. For example, we can ask if the variety is *unirational* meaning its points are parametrized by rational functions. This property ensures that many solutions to the defining equations can be easily written down.

Definition 1.2.1. Let X be a variety over k. We say that X is *unirational* (over k) if there is a dominant rational map $\mathbb{P}^n_k \dashrightarrow X$.

Unirationality can be viewed as a weaker notion of a polynomial system being "solvable" by rational functions where we do not require the parametrization by rational functions to give a precise 1-to-1 description of the set of solutions. In the 19th century it was understood that the Riemann-Hurwitz formula shows that – over the complex numbers – only genus 0 curves are unirational. From the function field perspective, Lüroth generalized this result to curves over any field. In modern notation he proved.

Theorem 1.2.2 (Lüroth, 1876)**.** *If X/k is a curve over a field of any characteristic then X is unirational iff X is rational. If moreover X is smooth and complete then $X \cong \mathbb{P}^1_k$.*

Lüroth asked if the higher-dimensional analog was true. In field-theoretic terms he asked if any finitely generated field extension K/k so that K is contained in a purely-transcendental extension $k(t_1, \ldots, t_n)$ then must K/k itself be purely transcendental. This question came to be known as the Lüroth problem. In more modern language, his problem asks whether a unirational variety of dimension n over k is always rational. The complex surface case, when $k = \mathbb{C}$ and $n = 2$ was also settled affirmatively in the 19th century by Castelnuovo.

Theorem 1.2.3 (Castelnuovo, 1896)**.** *Let $X/\mathbb{C}$ be a smooth projective surface. Then the following are equivalent,*

1. *X is unirational*
2. *X is rational*
3. *$p_2(X) := \dim_{\mathbb{C}} H^0(X, \omega_X^{\otimes 2}) = 0$ and $b_1(X) := \dim_{\mathbb{C}} H^1(X, \mathbb{C}) = 0$.*

Castelnuovo proved the equivalence by giving a purely numerical criterion – statement (3) – for rationality. He then showed that unirationality implies these numerical invariant vanish. The numerical invariants in question, the second *pluri-genus* p_2 and the first *Betti number*, fit into a more general framework of *obstructions* to unirationality (over $\mathbb{C}$) that apply in any dimension.

Definition 1.2.4. The m-th *pluri-genus* of X is the dimension of the space of sections of $\omega_X^{\otimes m}$,

$$p_m(X) := \dim_k H^0(X, \omega_X^{\otimes m}).$$

This is a birational invariant of smooth proper varieties. More generally, we can look at tensor forms i.e. sections of $(\Omega_X)^{\otimes m}$ and these can be pulled back along any rational map $f : X \dashrightarrow Y$ of obtain an obstruction to unirationality.

Proposition 1.2.5. *Let k be of characteristic zero and X a proper k-variety. Then if $\omega \in H^0(X, \Omega_X^{\otimes m})$ is a nonzero form for $m > 0$ then X is not unirational.*

Proof. Indeed, if $f : \mathbb{P}^n \dashrightarrow X$ is a unirational parametrization with $n = \dim X$ then f is generically étale. Therefore, on the locus $U \subset \mathbb{P}^n$ where f is a morphism, $f^*\omega$ is not zero identically. Furthermore, since X is proper, f is defined away from codimension ≥ 2 so $f^*\omega$ extends to a nonzer section of $H^0(\mathbb{P}^n, \Omega_{\mathbb{P}^n}^{\otimes m})$. No such section exists. □

It turns out to be vital that k has characteristic zero here as we will see in the subsequent section. The second invariant $b_1(X) := \dim_{\mathbb{C}} H^1(X, \mathbb{C})$ also fits into a larger story. The more refined obstruction is the fundamental group $\pi_1^{\text{top}}(X_{\mathbb{C}})$ or, in positive characteristic, the étale fundamental group $\pi_1^{\text{ét}}(X_{\bar{k}})$.

Theorem 1.2.6. *Let X be a unirational smooth projective variety over an algebraically closed field k.* find citation

1. *if k has characteristic zero then $\pi_1^{\text{ét}}(X) = 1$ and $\pi_1^{\text{top}}(X_{\mathbb{C}}) = 1$ for any $k \hookrightarrow \mathbb{C}$*
2. *if k has characteristic $p > 0$ then $\pi_1^{\text{ét}}(X)$ is a finite group of order coprime to p.*

In the subsequent section, we will see examples showing that the second statement is actual sharp. Therefore, both the conditions $p_2(X) = 0$ and $b_1(X) = 0$ fit into a larger story of *obstructions* to unirationality over $\mathbb{C}$. What turns out to be special to dimension 2 is that the vanishing of these obstructions $p_2 = 0$ and $b_1 = 0$ exactly characterizes rationality. Indeed, in dimension 3, a slew of unirational but non-rational examples were discovered in the year 1971 (some appearing in 1972). These examples are: any smooth cubic 3-fold by the work of Clemens and Girffiths [**?**], certain quartic 3-folds by the work of Iskovskikh and Manin [**?**], and certain Brauer–Severi varieties over rational surfaces by work of Artin and Mumford [**?**]. As illustrated by the complexity and variety of the those cited works, proving that a variety is not unirational is often quite subtle. For example, even in characteristic zero there is no known example of a rationally-connected variety that we can prove to be non-unirational.

1.3 The positive characteristic landscape and Shioda's conjecture (TODO)

The pullback argument in the proof of Proposition 1.2.5 fails completely in positive characteristic because of the existence of inseparable maps for which f^* is identically zero. If the map $f : \mathbb{P}^n \dashrightarrow X$ were separable then f^* would be injective just like in characteristic zero so the argument would go through. However, Zariski constructed examples of surfaces over $\mathbb{F}_p$ with an inseparable unirational parametrization but no separable one due to the existence of pluri-forms.

Example 1.3.1 (Zariski)**.** [**?**, §8] Consider a surface $X/\mathbb{F}_p$ defined as a smooth compactification of the affine surface

$$\{z^p = f(x, y)\} \subset \mathbb{A}^3_{\mathbb{F}_p}$$

for some $f \in \mathbb{F}_p[x, y]$. If we choose f generically of sufficiently large degree X will be of general type. However, notice that if we adjoin p^{th}-roots s, t of x and y then the equation becomes

$$z^p = f(s^p, t^p) = f(s, t)^p$$

therefore $k(X) \subset k(s, t)$ given by $z = f(s, t)$ and $x = s^p$ and $y = t^p$. This shows that X is always unirational via the inseparable map $\mathbb{A}^2 \dashrightarrow X$ given by

$$x \mapsto s^p, \quad y \mapsto t^p, \quad z \mapsto f(s, t).$$

This demonstrates that unirationality in positive characteristic is a phenomenon that persists even for quite "complex" varieties from the perspective of the usual classification of surfaces. This shows there are even fewer tools available to us in positive characteristic to prove that varieties are not unirational. The main goal of this thesis is to develop some new methods to accomplish this. The main ideas of this work are in conversation with two remarkable results from 1977: Shioda's classification of unirational Fermat surfaces and Bogomolov's boundedness theorem for curves on general type surfaces with $c_1^2 > c_2$. We will develop obstructions inspiried by the techniques of the later work and use them to answer questions asked in the former.

1.3.1 Shioda's Examples

Zariski's surfaces are cooked up precisely to be unirational in the specific characteristic p which is baked into the form of the equation. However, Shioda found an even more

remarkable example where a simple equation over $\mathbb{Z}$ can have infinitely many primes with unirational reduction mod p. Shioda showed the following by an inspired computation.

TODO, order of Shioda's examples, when do I explain the etale cohomology?

Theorem 1.3.2 (Shioda)**.** *Let F_n be the Fermat surface over $\mathbb{F}_p$ ($p > 2$) given by the equation*

$$\{X^n + Y^n + Z^n + W^n = 0\} \subset \mathbb{P}^3$$

Then F_n is unirational if and only if $p^\nu \equiv -1 \mod n$ for some ν.

This congruence condition is satisfied for all primes in certain arithmetic progressions and hence F_n is unirational at a set of primes of positive natural density by Dirichlet's theorem.

How does Shioda prove this theorem? For the "if" part, he writes down the unirational parametrization by an ingenious change of variables. To show the "only if," he needs an obstruction; this comes from étale cohomology.

Proposition 1.3.3. *Let X be a smooth projective unirational surface over $\mathbb{F}_q$. Then $\mathrm{Frob}_q \circlearrowright H^2_{\text{ét}}(X, \mathbb{Q}_\ell)$ has eigenvalues of the form $\zeta \cdot q$ for ζ a root of unity. We call this property supersingularity.*

Proof. Pass to an extension $\mathbb{F}_{q^n}$ such that the unirational parametrization and all points of indeterminacy are defined over $\mathbb{F}_{q^n}$. Resolving the indeterminacy locus, we get $\mathrm{Bl}_S\mathbb{P}^2 \to X$ where $S \subset \mathbb{P}^2$ is a finite set of points. Therefore $H^2_{\text{ét}}(X, \mathbb{Q}_\ell) \hookrightarrow H^2_{\text{ét}}(\mathrm{Bl}_S\mathbb{P}^2, \mathbb{Q}_\ell) = \mathbb{Q}_\ell(-1)^{1+\#S}$. Thus $\mathrm{Frob}_{q^n} = \mathrm{Frob}_q^n$ acts via q^n. □

Example 1.3.4. Let $G = \mathbb{Z}/5\mathbb{Z}$ act on F_5 by $\lambda \cdot [X : Y : Z : W] = [X : \lambda Y : \lambda^2 Z : \lambda^3 W]$. This action is free and hence $X = F_5/G$ is a smooth projective surface with $\pi_1(X) = \mathbb{Z}/5\mathbb{Z}$ called the Godeaux surface. Shioda's result shows that this is supersingular over infinitely many primes. However, this cannot happen over $\mathbb{C}$, since X has general type.

1.4 AGAIN

Theorem 1.4.1 (Shioda (1974), Shioda-Katsura (1979))**.** *Let X_n be the degree m Fermat surface $\{X^m + Y^m + Z^m + W^m = 0\}$ inside $\mathbb{P}^3_{\mathbb{F}_p}$ in characteristic p. The following are equivalent,*

1. *X_m is unirational over $\overline{\mathbb{F}_p}$*
2. *$-1 \in \langle p \rangle$ inside $(\mathbb{Z}/m\mathbb{Z})^\times$ i.e. there is ν such that $p^\nu \equiv -1 \mod m$*

3. *$H^2_{\text{ét}}(X_m, \mathbb{Q}_\ell)$ is spanned by algebraic cycles.*

In particular this occurs at a positive density set of primes. Let me say something about the proof but first we should talk about algebraic cycles and cohomology.

Definition 1.4.2. We say a surface X is *Shioda-supersingular* if $H^2_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell)$ is spanned by algebraic cycles.

Lemma 1.4.3. *If X is unirational it is (Shioda) supersingular.*

Proof. The rational map $f : \mathbb{P}^2 \dashrightarrow X$ can be resolved by blowing up $\mathbb{P}^2$ at finitely many points (possibly defined over k'/k) to get $\widetilde{f} : \mathrm{Bl}\mathbb{P}^2 \to X$. This dominant morphism induces a Galois-equivariant injection

$$H^2_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell) \hookrightarrow H^2_{\text{ét}}(\mathbb{P}^2, \mathbb{Q}_\ell)$$

Over k', the Galois-module $H^2_{\text{ét}}(\mathbb{P}^2, \mathbb{Q}_\ell)$ is spanned by algebraic cycles corresponding to the hyperplane and exceptional divisors. Pushing forward generators for the image, $H^2_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell)$ is also generated by algebraic cycles and, *a fortiori* X is supersingular. □

Based on these examples, he conjectured:

Conjecture (Shioda, 1977). *Let X be a smooth projective surface over an algebraically closed field k of characteristic p with $\pi_1^{\text{ét}}(X) = 1$. Then X is unirational if and only if it is Shioda supersingular.*

1.5 AGAIN AGAIN

This example gives a general type surface that is unirational in a particular characteristic p. However, there are even more disturbing examples of smooth projective general type surfaces over $\operatorname{Spec}\mathbb{Z}$ which are non-the-less unirational at a positive (arbitrarily close to 1) density set of primes. The first such example is due to Shioda.

Theorem 1.5.1 (Shioda (1974), Shioda-Katsura (1979))**.** *Let X_n be the degree m Fermat surface $\{X^m + Y^m + Z^m + W^m = 0\}$ inside $\mathbb{P}^3_{\mathbb{F}_p}$ in characteristic p. The following are equivalent,*

1. *X_m is unirational over $\overline{\mathbb{F}_p}$*
2. *$-1 \in \langle p \rangle$ inside $(\mathbb{Z}/m\mathbb{Z})^\times$ i.e. there is ν such that $p^\nu \equiv -1 \mod m$*
3. *$H^2_{\text{ét}}(X_m, \mathbb{Q}_\ell)$ is spanned by algebraic cycles.*

In particular this occurs at a positive density set of primes. Let me say something about the proof but first we should talk about algebraic cycles and cohomology.

First, recall that the cycle class map

$$\gamma : Z^1(X) \to H^2_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(1))$$

lands inside the $\mathbb{Q}_\ell$-subspace $H^2_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(1))^{\mathrm{Gal}(\bar{k}/k)}$ of Galois invariants. For $k = \mathbb{F}_q$, this is the subspace where Frob_p acts via $p \cdot \mathrm{id}$ (due to the presence of a Tate twist). The Tate conjecture predicts equality between this and the $\mathbb{Q}_\ell$-span of $\operatorname{im}\gamma$, the space of algebraic cycles.

Definition 1.5.2. We say a surface X is *Shioda-supersingular* if $H^2_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell)$ is spanned by algebraic cycles.

This implies X is *supersingular* meaning $H^2_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell(1))$ is entirely Galois-invariant after possibly some field extension k'/k where all cycles are defined. This is the same as saying that Frob_p^k for some k acts via $q^k \cdot \mathrm{id}$ or equivalently that Frob_p has eigenvalues all of the form $\zeta \cdot q$ where ζ is a root of unity. This last property is computationally checkable by point counting and the Weil conjectures. The Tate conjecture predicts that Shioda-supersingularity is equivalent to supersingularity. For all the surfaces I will discuss from now on, we know the Tate conjecture, so one should not worry at all about this distinction.

Lemma 1.5.3. *If X is unirational it is (Shioda) supersingular.*

Proof. The rational map $f : \mathbb{P}^2 \dashrightarrow X$ can be resolved by blowing up $\mathbb{P}^2$ at finitely many points (possibly defined over k'/k) to get $\widetilde{f} : \mathrm{Bl}\mathbb{P}^2 \to X$. This dominant morphism induces a Galois-equivariant injection

$$H^2_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell) \hookrightarrow H^2_{\text{ét}}(\mathbb{P}^2, \mathbb{Q}_\ell)$$

Over k', the Galois-module $H^2_{\text{ét}}(\mathbb{P}^2, \mathbb{Q}_\ell)$ is spanned by algebraic cycles corresponding to the hyperplane and exceptional divisors. Pushing forward generators for the image, $H^2_{\text{ét}}(X_{\bar{k}}, \mathbb{Q}_\ell)$ is also generated by algebraic cycles and, *a fortiori* X is supersingular. □

Proof. If $p^v \equiv -1 \mod m$ then Shioda wrote down an explicit change of variables and unirational parametrization $\mathbb{P}^2 \dashrightarrow X_m$. To go the other way, Shioda and Katsura proved that X_m is supersingular if and only if $-1 \in \langle p \rangle$ using the Weil conjectures and particularly Weil's early work on these conjectures working out point counting formulae in terms of Jacobi and Gauss sums. Therefore, if $-1 \notin \langle p \rangle$ then X_m cannot be unirational because otherwise X_m would be supersingular. □

Theorem 1.5.4 (Shioda, 1977)**.** *Let A be an abelian surface in characteristic > 2. Then the following are equivalent:*

1. *(A) is unirational*
2. *(A) is (Shioda) supersingular*
3. *A is Shioda supersingular*
4. *A is supersingular (equivalent to the usual definition: A is isogenous to $E \times E$ for E supersingular elliptic curve)*

He also worked out some examples with Delsarte surfaces. Based on these examples, he conjectured:

Conjecture (Shioda, 1977). *Let X be a smooth projective surface over an algebraically closed field k of characteristic p with $\pi_1^{\text{ét}}(X) = 1$. Then X is unirational if and only if it is Shioda supersingular.*

Remark 1.5.5. Shioda knew counterexamples if we relaxed the simply-connected condition to $\pi_1^{\text{ét}}(X)$ being finite of order coprime to p. Indeed, in that case X is unirational if and only if its universal cover $\widetilde{X} \to X$ is but $\widetilde{X}$ may have additional H^2 not spanned by algebraic cycles. One should, in this case, consider the conjecture for $\widetilde{X}$ not for X.

For an explicit example, Shioda used the Godeaux surface constructed as a free μ_5 quotient of the Fermat X_5 of degree 5 which is supersingular in all characteristics but unirational if and only if X_5 is i.e. when $p^v \equiv -1 \mod 5$.

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Remark 1.5.6. One reason, known to Shioda, to doubt the validity of this conjecture is the failure of a particular characteristic zero analog. The divisibility properties of the Frob-eigenvalues is analogous to properties of the Hodge structure. In particular, we think of Frob-eigenvalues on H^2 not divisible by p as analogous to classes in $H^{2,0} \oplus H^{0,2}$. Therefore, supersingularity should correspond to having a trivial Hodge structure $H^2 = H^{1,1}$ so that H^2 is spanned by algebraic cycles.

One can ask if complex surfaces X with $\pi_1(X) = 1$ and $H^2 = H^{1,1}$ must be unirational (hence rational). This is false. There are examples due to Dolgachev of elliptic surfaces. Further examples due to an example of Barlow of a general type simply-connected surface with Hodge diamond,

$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 0 & & 9 & & 0 \\ & 0 & & 0 & \\ & & 1 & & \end{array}$$

However, the existence of inseparable unirational parametrization makes this analogy less convincing. It is a very interesting question to ask if the reduction of a Barlow surface satisfies Shioda's conjecture. I suspect not.

1.6 Addendum (REMOVE)

Lemma 1.6.1. *Let $f : X \to Y$ be surjection of smooth proper varieties. Then $f^* : H^i_{\text{ét}}(Y, \mathbb{Q}_\ell) \hookrightarrow H^i_{\text{ét}}(X, \mathbb{Q}_\ell)$ is injective.*

Proof. Poincaré duality defines a pushforward map $f_* = D^{-1} \circ (f_*)^\vee \circ D$ where $D(\alpha) = \operatorname{tr}(\alpha \smile -)$. If f is generically finite, it suffices to show that $f_* f^* = \deg f$ since this is invertible on $\mathbb{Q}_\ell$. Indeed,

$$D(f_* f^* \alpha) = \operatorname{tr}(f^* \alpha \smile f^* -) = \operatorname{tr} f^*(\alpha \smile -) = (\deg f) D(\alpha)$$

because $f^* = \deg f$ on top cohomology. If f is not generically finite then we may take a generic multisection $Z \subset X \to Y$ and then $(f \circ \iota)^*$ is injective so f^* is injective. □

1.6.1 Shioda's Example

Shioda showed the following by a remarkable computation.

Theorem 1.6.2 (Shioda)**.** *Let F_n be the Fermat surface over $\mathbb{F}_p$ ($p > 2$) given by the equation*

$$\{X^n + Y^n + Z^n + W^n = 0\} \subset \mathbb{P}^3$$

Then F_n is unirational if and only if $p^\nu \equiv -1 \mod n$ for some ν.

Example 1.6.3. Let $G = \mathbb{Z}/5\mathbb{Z}$ act on F_5 by $\lambda \cdot [X : Y : Z : W] = [X : \lambda Y : \lambda^2 Z : \lambda^3 W]$. This action is free and hence $X = F_5/G$ is a smooth projective surface with $\pi_1(X) = \mathbb{Z}/5\mathbb{Z}$ called the Godeaux surface. Shioda's result shows that this is supersingular over infinitely many primes. However, this cannot happen over $\mathbb{C}$, since X has general type.

How does Shioda prove this theorem? For the "if" part, he writes down the unirational parametrization by an ingenious change of variables. To show the "only if," he needs an obstruction; this comes from étale cohomology.

Proposition 1.6.4. *Let X be a smooth projective unirational surface over $\mathbb{F}_q$. Then $\mathrm{Frob}_q \circlearrowright H^2_{\text{ét}}(X, \mathbb{Q}_\ell)$ has eigenvalues of the form $\zeta \cdot q$ for ζ a root of unity. We call this property supersingularity.*

Proof. Pass to an extension $\mathbb{F}_{q^n}$ such that the unirational parametrization and all points of indeterminacy are defined over $\mathbb{F}_{q^n}$. Resolving the indeterminacy locus, we get $\mathrm{Bl}_S\mathbb{P}^2 \to X$ where $S \subset \mathbb{P}^2$ is a finite set of points. Therefore $H^2_{\text{ét}}(X, \mathbb{Q}_\ell) \hookrightarrow H^2_{\text{ét}}(\mathrm{Bl}_S\mathbb{P}^2, \mathbb{Q}_\ell) = \mathbb{Q}_\ell(-1)^{1+\#S}$. Thus $\mathrm{Frob}_{q^n} = \mathrm{Frob}_q^n$ acts via q^n. □

Shioda then computes that F_n is supersingular exactly when $p^\nu \equiv -1 \mod n$ for some ν and this covers all the cases his construction works to show F_n is moreover unirational. This, and a handful of other examples, leads him to make the following conjecture.

Conjecture 1.6.5 (Shioda)**.** *Let X be a smooth projective surface over $\mathbb{F}_q$. Suppose that*

1. $\pi_1(X) = 1$
2. *X is supersingular*

then X is unirational.

Remark 1.6.6. One might ask why Shioda does not simply require $\pi_1(X)$ to be finite, as suggested by Serre's theorem and the Godeaux example. Such a conjecture would be false: the Godeaux surface is supersingular in every characteristic of good reduction but is unirational only in those characteristics where F_5 is unirational.

The first goal of this thesis is to exhibit a counterexample to Shioda's conjecture. In fact, we will do more we will show examples that are not covered by genus 0 or genus 1 curves. A variety covered by rational curves is called *uniruled*. Precisely, X is uniruled if there is a dominant rational map $f : W \times \mathbb{P}^1 \dashrightarrow X$ such that f does not factor through $W \times \mathbb{P}^1 \to W$. In Chapter **??** we will prove the following theorem by exhibting such a surface X contradicting the Shioda conjecture.

Theorem 1.6.7. ?? *There exists a smooth projective surface over $\overline{\mathbb{F}_p}$ with the following properties:* citation

1. *$H^2_{\text{ét}}(X, \mathbb{Q}_\ell)$ is spanned by algebraic cycles*
2. $\pi_1^{\text{ét}}(X) = 1$
3. *X is not unirational, in fact X is not uniruled or covered by genus 1 curves.*

1.7 Unirationality of Moduli Spaces (TODO)

Another motivation to go beyond the obstructions arising from étale cohomology is to prove the non-uniruledness of moduli spaces in characteristic p. In characteristic 0, pluri-forms have a long and storied history establishing non-uniruledness of moduli spaces such as $M_{g,n}$ but are insufficient to make any headway in positive characteristic because of the issues we have seen regarding inseparable map.

The (non)-unirationality of moduli spaces is particularly interesting because the abstract birational geometry has implications for the behavior of the generic objects parametrized in the universal family. A moduli space is unirational exactly when the generic object it parametrizes can be "written down" by an explicit algebraic recipe. Explicitly, this means there is a versal family over a rational variety whose generic point gives the generic object writen down in terms of "free parameters", the transcendental coordinates $t_1, \ldots, t_n$ on the fraction field $k(t_1, \ldots, t_n)$.

Example 1.7.1. Every genus 2 curve is hyperelliptic meaning there is a degree 2 map $C \to \mathbb{P}^1$ ramified over exactly 6 branch points in $\mathbb{P}^1$. This means it can be written in the form

$$y^2 = x(x-1)(x-\lambda_1)(x-\lambda_2)(x-\lambda_3)$$

choosing coordinates on $\mathbb{P}^1$ so that the branch points occur at $0, 1, \infty, \lambda_1, \lambda_2, \lambda_3$. Since the parameters $\lambda_1, \lambda_2, \lambda_3$ are freely varying in an open set of $\mathbb{A}^3$, this shows that M_2 is 3-dimensional and unirational.

For small values of g, the moduli space of curves M_g is known to be unirational. However, A celebrated result of Harris and Mumford proves that the moduli space of curves of genus g is of general type for $g > 23$. In characteristic zero, this implies that M_g is not unirational or even uniruled. More generally, the following result shows that only finitely many $M_{g,n}$ are uniruled.

Theorem 1.7.2 (Harris–Mumford (1982) + many others)**.** *$\kappa(M_{g,n}) \geq 0$ for $g > 22$ or $n > 15$. Hence, in these cases, $M_{g,n_{\mathbb{C}}}$ is not uniruled.* CITE

A particularly nice case is $g = 1$ and $n \geq 11$ where the cusp form Δ proves $\kappa(M_{1,n}) \geq 0$. Sawin used more delicate arguments using the Galois representation on étale cohomology as an obstruction to prove certain cases of the non-uniruledness of $M_{1,n}$ in positive characteristic.

Theorem 1.7.3 (Sawin 2019)**.** *$(M_{1,n})_{\overline{\mathbb{F}}_p}$ is not uniruled as long as $n \geq p \geq 11$.* CITE

Even in this case, for fixed n, we do not know if $M_{1,n}$ is unirational or uniruled in all but finitely many characteristics. To my knowledge, we know next to nothing about the uniruledness or unirationality of $M_{g,n}$ for $g > 1$ in positive characteristic.

The difficulty of using techniques based on étale cohomology to prove uniform obstruction results over different primes partially inspired the investigation of the non-cohomological obstructions we consider in this thesis. Our techniques do not yet apply to the moduli spaces of curves $M_{g,n}$. However, we are able to apply them to Hilbert modular varieties to obtain uniform obstruction results.

Theorem 1.7.4. *Let $Y_0^F(N)$ be the Hilbert modular variety at level $\Gamma_0(N)$ associated to a totally field F. Then either N or the discriminant of F are sufficiently large, the reduction $Y_0^F(N)_{\overline{\mathbb{F}}_p}$ mod p is not uniruled or covered by genus 1 curves for all primes p not dividing N.* CITE

In particular, the obstruction applies to all but finitely many characteristics simultaneously.

1.8 Hyperbolicity (TODO)

Many of the techniques in this thesis were motivated by methods used to study hyperbolicity of complex manifolds. Our investigations go beyond rationality questions to problems more directly connected to hyperbolicity. Here we review some of the history, definitions, and main conjectures of this area.

1.8.1 The Kobayashi Pseudodistance

In complex analysis, hyperbolic spaces such as bounded symmetric domains often carry a canonical *Bergman metric* characterized by the remarkable property that every biholomorphism is an isometry. For example, the complex hyperbolic space $\mathbb{H}^n_{\mathbb{C}}$ which is the unit ball $\mathbb{B}^n \subset \mathbb{C}^n$ equipped with a certain metric of constant negative holomorphic bisectional curvature satisfies this property. The miracle is that the complex structure determines a metric, unique up to all holomorphic reparametrizations, whose metric geometry encodes the complex geometry. For positively curved spaces such as $\mathbb{P}^n$ there can be no such biholomorphism invariant metric. This is because maps like the homothety $z \mapsto \lambda z$ on $\mathbb{P}^1$ are fundamentally non-area preserving conformal transformations. More precisely, the biholomorphism group PGL_{n+1} of $\mathbb{P}^n$ is noncompact and has nontrivial stabilizers of a point and its tangent space; hence it cannot be contained in any group of isometries.

However, motivated by the Schwarz lemma (1869) and its generalization by Ahlfors (1938), Kobayashi discovered there is always a canonical *pseudo-metric* determined only by the complex structure which is invariant under all biholomorphisms. Here let $\mathbb{D} \subset \mathbb{C}$ be the unit disk and ρ the Poincaré metric on $\mathbb{D}$.

Definition 1.8.1. Let X be a complex manifold. The *Kobayashi pseudodistance* is the pseudometric defined as follows:

$$d_X(p, q) = \inf_{\alpha} \ell(\alpha)$$

where α is a chain of holomorphic disks $f_i : \mathbb{D} \to X$ and points $p = p_0, p_1, \ldots, p_k = q$ of X and pairs $a_1, b_1, \ldots, a_k, b_k \in \mathbb{D}$ such that,

$$f_i(a_i) = p_{i-1} \quad f_i(b_i) = p_i$$

and the length $\ell(\alpha)$ of the chain is defined as,

$$\ell(\alpha) := \rho(a_1, b_1) + \cdots + \rho(a_k, b_k).$$

Example 1.8.2. Let $X = \mathbb{C}$ then $d_X = 0$. Indeed, by choosing larger and larger discs containing p, q their pullback to the unit disk is then closer and closer to the origin and hence has vanishing Poincaré distance.

Example 1.8.3. Recall the Schwarz–Pick lemma says that any holomorphic map $f : \mathbb{D} \to \mathbb{D}$ is a contraction for the Poincare metric. Therefore, $d_{\mathbb{D}} = \rho$.

Definition 1.8.4. We say that a complex manifold X is *(Kobayashi)-hyperbolic* if d_X is an honest metric meaning if $d_X(x, y) = 0$ then $x = y$.

The Kobayashi pseudodistance is not only intrinsic to X and hence preserved by all biholomorphic reparametrizations, it is actually length non-decreasing (i.e. *short*) for all holomorphic maps.

Lemma 1.8.5. *Let $f : X \to Y$ be holomorphic. Then $d_Y(f(x), f(y)) \leq d_X(x, y)$*

Proof. Indeed, choosing any chain of disks $g_i : \mathbb{D} \to X$ computing $d_X(x, y)$ we see that $f \circ g_i$ is a chain of disks connecting $f(x)$ and $f(y)$ of the same length. Therefore,

$$d_Y(f(x), f(y)) = \inf_{\alpha} \ell(\alpha) \leq d_X(x, y)$$

□

Corollary 1.8.6. *If $f : \mathbb{C} \to X$ is an entire curve then for $x, y \in f(\mathbb{C})$ we have $d_X(x, y) = 0$ meaning if f is nonconstant then d_X is degenerate along the image of f.*

Proof. Indeed, let $z_1, z_2 \in \mathbb{C}$ map to x, y respectively. Then,

$$d_X(x, y) \leq d_{\mathbb{C}}(z_1, z_2) = 0$$

□

Brody provided a converse to this result.

Theorem 1.8.7 (Brody)**.** *Let X be a compact complex manifold. If d_X is degenerate then there exists a nonconstant entire curve $f : \mathbb{C} \to X$.*

Definition 1.8.8. A complex manifold is *Brody-hyperbolic* if there is no nonconstant entire map $f : \mathbb{C} \to X$.

In particular, if X is Brody hyperbolic then X does not contain any rational or elliptic curves. We will return to this observation in Demailly's notion of *algebraically hyperbolic*. Therefore, as a consequence of Brody's theorem we obtain the following equivalence.

Theorem 1.8.9 (Brody)**.** *Let X be a compact complex manifold. Then X is Kobayashi hyperbolic if and only if it is Brody hyperbolic.*

Therefore, we will call compact manifolds with this property just "hyperbolic" or "analytically hyperbolic" for emphasis. In the noncompact case, these two notions do differ [Bar80].

1.8.2 Algebraic Hyperbolicity

The importance of Brody's reinterpretation of hyperbolicity for compact manifolds is that it resembles a form that more closely touches algebraic geometry. This connection is made concrete by another redefinition due to Demailly which is conjecturally equivalent to hyperbolicity for smooth projective varieties.

Definition 1.8.10. Let X be a projective variety and H an ample divisor. We say X is *algebraically hyperbolic* if there exists $\epsilon > 0$ such that for every complete integral curve $C \subset X$ we have,

$$2g(C) - 2 \geq \epsilon \deg_H C$$

where $g(C)$ is the geometric genus of C.

Demailly proved that algebraic hyperbolicity is at least a weaker notion than analytic hyperbolicity in the projective setting.

Theorem 1.8.11 (Demailly)**.** *Let X be a smooth projective variety. Then*

$$X \text{ is analytically hyperbolic} \implies X \text{ is algebraically hyperbolic}.$$

Moreover, he conjectured the converse to hold as well.

Conjecture 1.8.12 (Demailly)**.** *Let X be a smooth projective variety. Then*

$$X \text{ is analytically hyperbolic} \implies X \text{ is algebraically hyperbolic}.$$

Algebraic hyperbolicity, being amenable to attack via techniques in algebraic geometry, is often easier to study. For example, the famous conjecture of Kobayashi predicting the hyperbolicity of general hypersurfaces remains open.

Conjecture 1.8.13 (Kobayashi)**.** *Let $X \subset \mathbb{P}^{n+1}$ be a very general hypersurface of degree $d \geq 2n+1$. Then X is hyperbolic.*

After a considerable effort by many authors, the conjecture is now known for $d \geq d_n$ where $d_n \sim n^4$ [BK24]. However, its algebraic variant is known almost completely [Voi96, CR19, Yeo25] by much more elementary means.

1.8.3 Green-Griffiths-Lang conjecture

It is expected that general type varieties are *close* to being hyperbolic in the following precise sense. We say that a projective variety X is *algebraically quasi-hyperbolic* if the images of all genus 0 and genus 1 curves lie in a proper Zariski closed subset $Z \subsetneq X$. In characteristic zero, this is the algebraic analog of a weakening of Brody hyperbolicity: we say that X is *quasi-hyperbolic* if the images of all nonconstant entire curves $f : \mathbb{C} \to X$ lie in a proper Zariski closed subset. The Green-Griffiths-Lang conjecture predicts this is always true for varieties of general type.

Conjecture 1.8.14 (Green–Griffiths–Lang)**.** *Let $X/\mathbb{C}$ be a smooth projective variety of general type. Then X is quasi-hyperbolic i.e. there is a proper Zariski closed subset $Z \subsetneq X$ containing the images of all nonconstant entire curves $f : \mathbb{C} \to X$.*

The minimum such Z is often called the *exceptional set* of X. The remarkable feature of this conjecture is that we expect every entire curve to be *algebraically degenerate* meaning

it satisfies a polynomial relation (i.e. it lives in Z). Moreover, we expect that these relations can be chosen independently of the map!

The Green–Griffiths–Lang conjecture is highly resonant with another wide open problem: the Bombieri–Lang conjecture predicting some control on the locus of rational points on general type varieties.

Conjecture 1.8.15 (Bombieri–Lang)**.** *Let X/K be a smooth projective variety over a number field K. Assume X is general type. Then there is a proper Zariski closed subset $Z \subsetneq X$ such that for each finite extension K'/K the set of K'-points outside Z is finite.*

An even more ambitious claim is that we should be able to take Z in the Bombieri-Lang conjecture to be the exceptional set of $X_{\mathbb{C}}$. This conjecture would imply a deep connection between the hyperbolic geometry of X and its distribution of rational points. To the author, no enduring mystery in mathematics is more compelling than understanding the nature of this connection. Unfortunately, we will have nothing to say about it in this thesis.

The strategy developed by Bogomolov, Green, Griffiths, Demailly and many others towards the Green–Griffiths–Lang conjecture is to produce algebraic differential equations that entire curves must satisfy. We can think of a first-order differential equation as a closed subspace of the tangent bundle of X. If we could produce enough differential equations so that the intersection of these loci inside the tangent bundle (or appropriate higher-order jet spaces) does not dominate X then we win.

Unfortunately, in many cases of interest known to be hyperbolic, this strategy fails because there are provably not enough algebraic differential equations to cut out a proper subvariety of X. For example, this occurs for products of hyperbolic curves and most Hilbert modular surfaces [DR15]. These are exactly the types of objects we need to study in this thesis, so a more sophisticated approach is needed.

1.8.4 Bogomolov's Boundedness Theorem (TODO)

The first real progress towards the Green–Griffiths–Lang conjecture is Bogomolov's celebrated boundedness theorem for complex surfaces of general type satisfying $c_1^2 > c_2$. Bogomolov's remarkable proof also provides a route around the roadblock mentioned above.

Theorem 1.8.16 (Bogomolov, 1977)**.** *Let X be a smooth projective surface with $s_2(X) = c_1(X)^2 - c_2(X) > 0$. Then the curves on X with fixed genus form a bounded family. In particular, X has finitely many genus 0 and genus 1 curves i.e. it is algebraically quasi-hyperbolic.*

discuss how Bogomolov fixes the problem above

get the bogomolov discussion from obstructions chapter?

this is not the optimal version of McQuillan YEAR?

Theorem 1.8.17 (McQuillan)**.** *If X is a smooth projective surface with $s_2(X) = c_1(X)^2 - c_2(X) > 0$ and X has no genus 0 or genus 1 curves, then X is hyperbolic.*

Ingredients in Bogomolov's proof:

1. existence of a symmetric differential form $\omega \in H^0(X, S^m\Omega_X)$
   (a) stability of the cotangent bundle
   (b) Bogomolov vanishing theorem
2. construction of foliations
3. Jouanolou's theorem on algebraic leaves of foliations

1.8.5 Positive Characteristic

The situation regarding hyperbolicity is rather different in positive characteristic. The analog of the Green–Griffiths–Lang conjecture fails spectacularly. In fact, we have already seen it fails maximally since a general type variety may be uniruled even unirational! Likewise, in chapter § we will construct examples of smooth projective surfaces of general type in positive characteristic that are provably not covered by genus 0 or genus 1 curves yet are nevertheless not algebraically quasi-hyperbolic. Explicitly, such a surface contains an infinite set of genus 0 or genus 1 curves that are "discrete" in the sense that none move in families and whose degrees grow without bound. In the same section, we will give a new criterion for algebraic quasi-hyperbolicity in positive characteristic. Then we use this criterion to show that analogous constructions to the above lead to surfaces that *are* algebraically quasi-hyperbolic showing the subtlety of this property in positive characteristic.

chapter:obstructions

sec:quasi-hyperbolicity

The techniques we develop to control the images of genus 0 and 1 curves in positive characteristic parallel, and are largely inspired by, the techniques mentioned above employing algebraic differential equations towards the Green-Griffiths-Lang conjecture or complex hyperbolicity. The Green–Griffiths strategy of building enough algebraic differential equations on X to force any genus ≤ 1 curve to be algebraically degenerate works just as well in positive characteristic using Hasse-Schmidt derivations [Cad26]. However, techniques based on foliations to push beyond the Green–Griffiths locus, such as in Bogomolov's argument, encounter major new difficulties in characteristic $p > 0$.

The main difficulty is that solutions to differential equations behave quite differently in positive characteristic, losing much of their rigidity. In particular, there is no uniqueness

theorem for solutions to ODEs with prescribed boundary conditions due to the basic fact that both 1 and x^p satisfy $f'(x) = 0$ in characteristic p. However, the solutions to differential equations are parametrized by subvarieties of a certain inseparable quotient (see chapter **??** Lemma **??**). This weak description of the solutions is enough to jump-start our technique. Specifically, we try to force the curves on the inseparable quotient to themselves satisfy additional algebraic differential equations. If we can ensure this happens *ad infinitum* then we reach a contradiction by infinite descent (or equivalently by the fact that an algebraic map has finite degree). Another way to view our technique is to build a mechanism to force higher and higher derivatives of the map to be degenerate. A toy model for what happens is as follows: if f is a polynomial whose divided derivatives $\frac{1}{n!}\frac{\mathrm{d}^n}{\mathrm{d}x^n}f = 0$ all vanish, then f is a constant. Of course, this trivially holds equally well in positive characteristic. In the main applications of this text to the Shioda conjecture and Hilbert modular varieties, we achieve this control over the higher order derivatives of a map $f : C \to X$ in a fairly roundabout way using the inseparable quotients. A more direct method to force arbitrarily high derivatives of a map $f : C \to X$ to vanish is to find algebraic differential equations of arbitrarily high order that f must satisfy in such a way as to impose nondegenerate conditions on the higher (divided) derivatives $f^{(n)}$. This version using higher-order differential equations arising from jet bundles is explored in chapter **??**.

1.9 Outline of Chapters (TODO)

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